THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department of Electronic and Computer Engineering
ELEC 1100

Laboratory 2: Kirchhoff's Laws (5%)

A) Objectives:

- To get familiar with circuit diagrams and circuit construction on a breadboard.
- To gain experience in measuring resistance, voltage, and current using the digital multimeter.
- To verify Kirchhoff's voltage and current laws.

B) Equipment:

- DC power supply, digital multi-meter, resistors.

C) Experiment Procedures:

- **Review all the techniques in connecting components and measurements in Lab1.**

Experiment 1: Study on the Equivalent Resistance (~10 mins)

Step 1: For the given resistor networks as shown below, determine the equivalent (total) resistance R_{eq} for each of them by calculation.

Q1: Calculate for R_{eq} and complete the table in the summary sheet. Show your calculation steps.

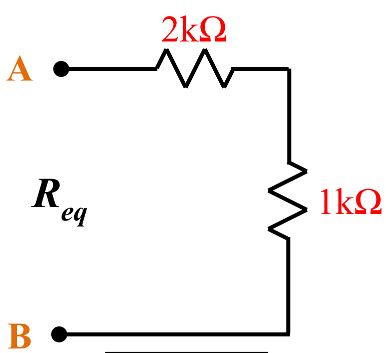


Figure 1

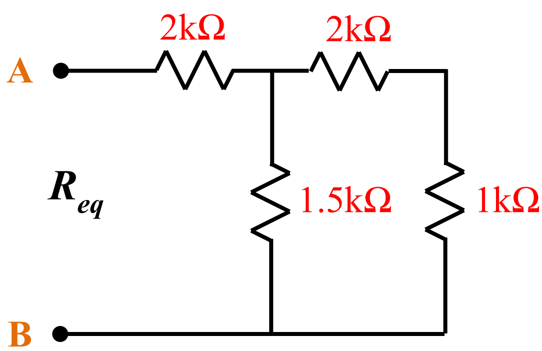


Figure 2

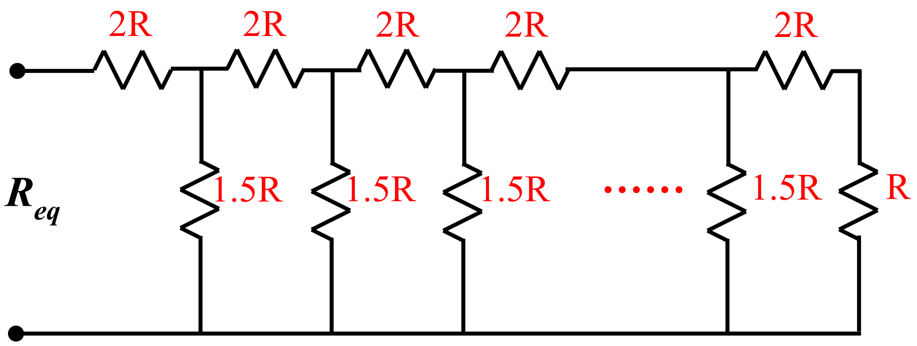
Step 2: Use the multimeter (switch to the resistance mode "Ω") to find the resistors needed and construct the circuits in Figure 1 and Figure 2 on your breadboard.

Step 3: For each of the two circuits above, use the multimeter to measure the resistance between **nodes A and B** to determine the value of R_{eq}.

Q2: Record the value of R_{eq} displayed on the screen (including the unit Ω/kΩ) to fill in the table in the summary sheet. This value should match your calculation in Q1.

******TA Check 1: Show the measurement of R_{eq} to your TA.**

Q3: Based on what you got above, analyze a resistor network with infinite modules (R, 1.5R, 2R) as below. Prove that $R_{eq} = 3R$. Show clearly the calculation steps.

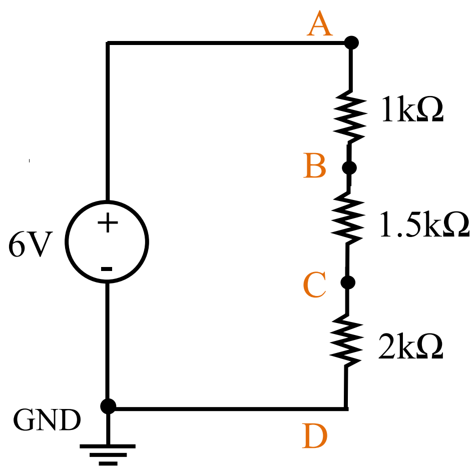


Experiment 2: Verify Kirchhoff's Voltage Law (~10 mins)

In this experiment, we use the measured voltages to verify Kirchhoff's voltage law (**KVL**), which indicates that the algebraic sum of the **voltages** around any closed path in a given circuit is always zero.

Step 1: Generate 6V from the DC power supply.

Step 2: Use the multimeter to find the resistors needed and construct the below circuit on the breadboard.



Step 3: Use the multimeter (switch to the voltage mode "V") to measure the voltage at node **A** relative to the power supply ground **GND** (i.e. connect the black lead of the multimeter to node D) to determine V_{AD}.

Step 4: Now measure the voltage at node **A** relative to node **B** (i.e. connect the black lead of the multimeter to node B) to determine V_{AB}.

Step 5: Now measure the voltage at node **B** relative to node **C** (i.e. connect the black lead of the multimeter to node C) to determine V_{BC}.

Step 6: Now measure the voltage at node **C** relative to the power supply ground **GND** (i.e. connect the black lead of the multimeter to node D) to determine V_{CD}.

Step 7: Now measure the voltage at node **D** relative to node **A** (i.e. connect the black lead of the multimeter to node A) to determine V_{DA}. Notice the voltage direction. You should have V_{DA} = -V_{AD}.

Q4: Record the measurement of voltages V_{AD}, V_{AB}, V_{BC}, V_{CD}, and V_{DA} to fill in the table in the summary sheet.

Q5: Complete the table to show that the sum of the voltages ACROSS all of the components (including the power source) in a loop complies with Kirchhoff's voltage law (**KVL**).

$$V_{AD} = V_{AB} + V_{BC} + V_{CD}$$

$$V_{AB} + V_{BC} + V_{CD} + V_{DA} = 0$$

******TA Check 2: Show your TA the complete table in the summary sheet.**

<u>Notes</u>:

1. The reference node in a circuit can be anywhere you choose in a real circuit, therefore the voltage at a node may change depending on your reference.

2. In this course, the **negative terminal** of a power supply is normally taken to be the reference node where the voltage = 0V. So in a circuit drawing, conceptually, the symbol 

is used to indicate the reference node.

3. 

is **NOT** referring to the **GND terminal** of a real power supply.

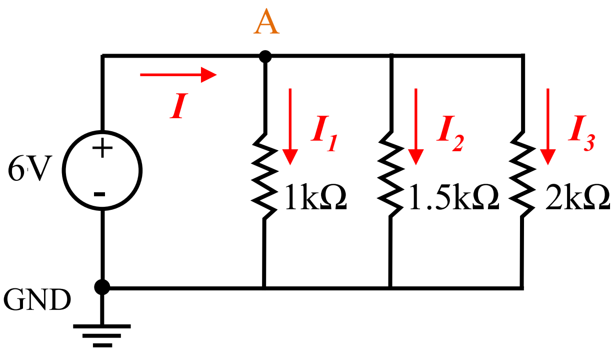
Experiment 3: Verify Kirchhoff's Current Law (~20 mins)

In this experiment, we use the measured currents to verify Kirchhoff's current law (**KCL**), which indicates that the algebraic sum of the **currents** entering a node equal to the sum of the currents leaving the same node.

In the circuit below, the voltage across any parallel resistors is always the same. Current through any resistor can be determined by using **Ohm's law**.

$$I = \frac{V}{R} = \frac{Voltage\ across\ a\ resistor}{Resistance}$$

Q6: Calculate for branch currents I_1, I_2, and I_3 to fill in the table in the summary sheet.



Step 1: Generate 6V from the DC power supply. Use the multimeter to find the resistors needed and construct the circuit above on your breadboard.

Now follow the below steps to measure the total current $\boldsymbol{I}$ using the multimeter.

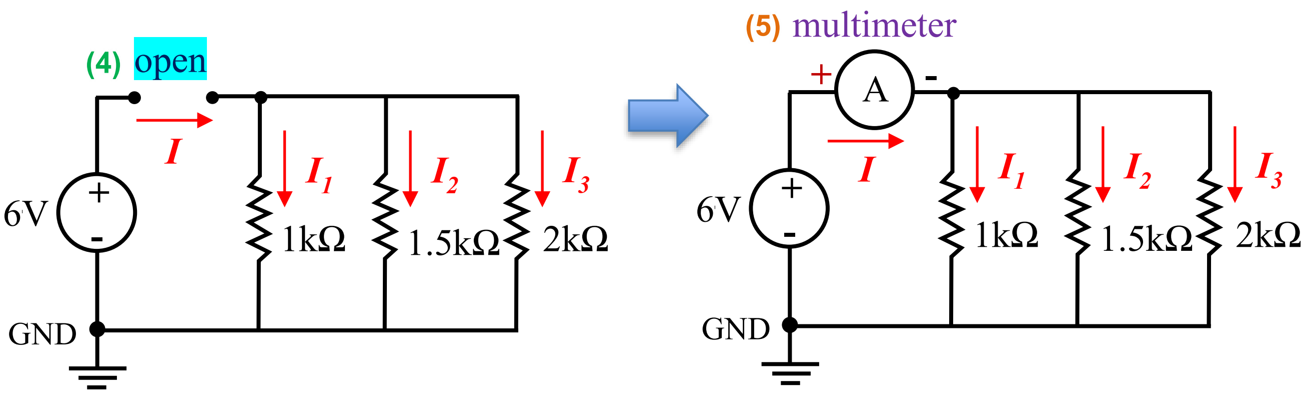
Step 2: On the multimeter, switch to the small current mode $\boxed{\overset{\simeq}{\text{mA}}}$.

Step 3: Plug in the multimeter probes to the input terminals as shown. Remember to match the colors for the common convention.

For measuring current, the multimeter **MUST** be connected <u>in series</u> with the component(s) at which the current is going through.

Step 4: Rewire the circuit on your breadboard to leave an open circuit as below, to ensure the multimeter could be connected <u>in series</u>.

Step 5: Connect the probe leads from the multimeter to measure the total current $\boldsymbol{I}$.



****** TA Check 3: Demo the measurement of current $\boldsymbol{I}$ to your TA.**

Q7: Record the value $\boldsymbol{I}$ displayed on the screen, including the unit "mA". Complete the table in the summary sheet to show that the sum of currents complies with Kirchhoff's current law (**KCL**).

$$I = I_1 + I_2 + I_3$$

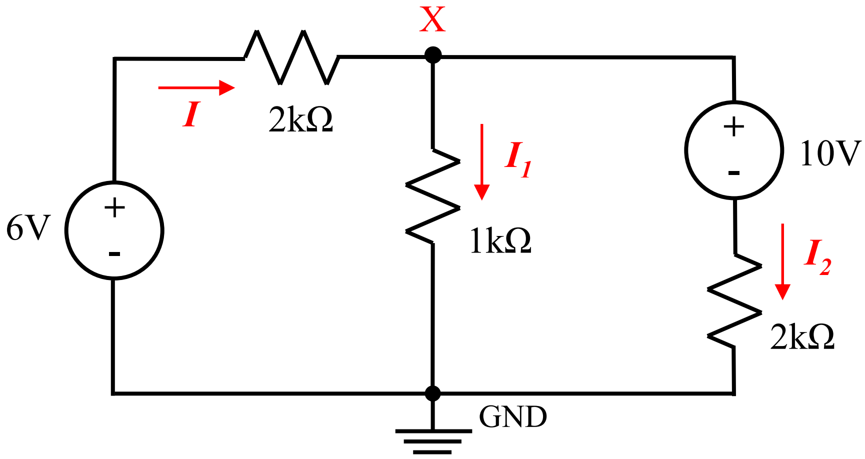
CAUTION

When using the multimeter for the current, "mA" terminals is in essence a short circuit and **MUST** be <u>in series</u> with the components (as a wire) through which current is being measured. NEVER connect "mA" terminals in parallel with any components you are trying to measure. An error in the connection could seriously damage the multimeter "mA" channels and the other circuit components.

Experiment 4: Study on the Two Power Sources Circuit (~30 mins)

In this experiment, we study a complex circuit that has two DC power sources.

Q8: Use **KCL** to solve the currents ***I***, I_1, and I_2 in the circuit below, then determine the voltage at node X. Show your calculation steps. Notice the unit and direction of each value.



Now we construct the circuit on the breadboard to verify the calculated results.

Step 1: Switch off the DC power supply and turn all the knobs fully anticlockwise.

Step 2: Construct the circuit on the breadboard. Connect the Slave Channel of the power supply to 6V of the circuit, and the Master Channel to 10V.

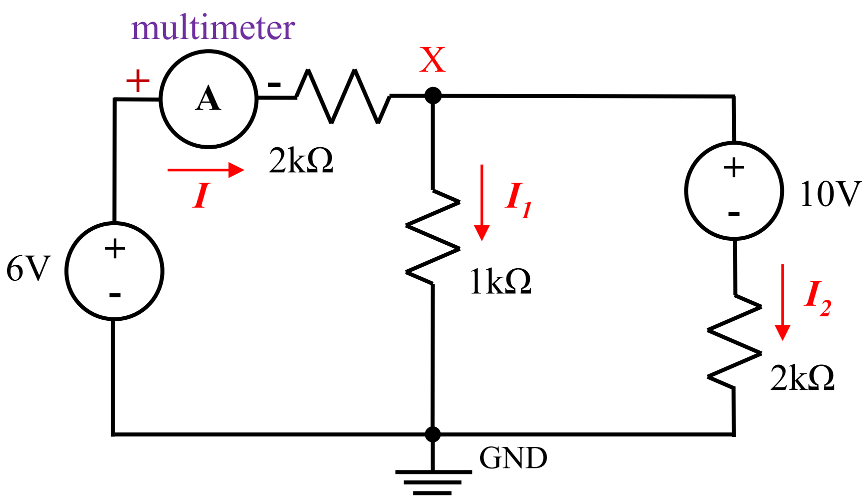
Step 3: Switch on the power supply. Set the current limits for both channels as in **Lab1**.

Step 4: Set Slave Channel to 6V. Set Master Channel to 10V.

Step 5: Use the multimeter "V" mode to measure the voltage at node X.

Q9: Record the value of V_X from the multimeter to fill in the table in the summary sheet. This value should match your calculated V_X in Q8.

Step 6: Rewire the circuit as shown in the figure below to connect the multimeter <u>in series</u> with the top 2kΩ resistor for measuring the total current ***I*** in the circuit.



Step 7: Change to use the multimeter "mA" mode as in **Experiment 3**.

****** TA Check 4: Demo the measurement of current *I* to your TA.**

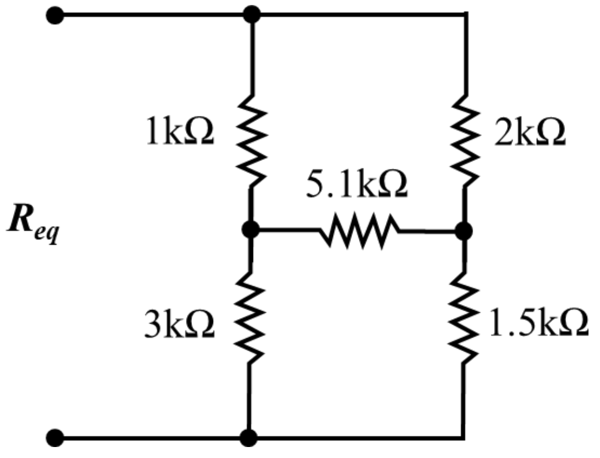
Q10: Record the value of I from the multimeter (including the unit "mA") to fill in the table in the summary sheet. This value should match your calculated I in Q8.

Experiment 5: Study on a Complex Circuit Network (~30 mins)

In this experiment, we study a complex 5-resistors circuit network.

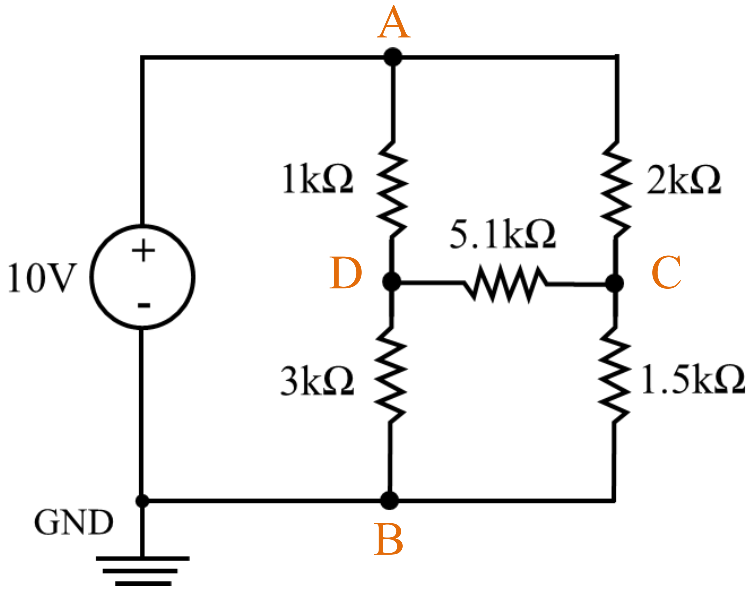
Step 1: Construct the below resistor network on the breadboard.

Step 2: Use the multimeter "Ω" mode to measure the equivalent (total) resistance R_{eq}.



Q11: Record the value of R_{eq} from the multimeter (including the unit Ω/kΩ) to fill in the table in the summary sheet.

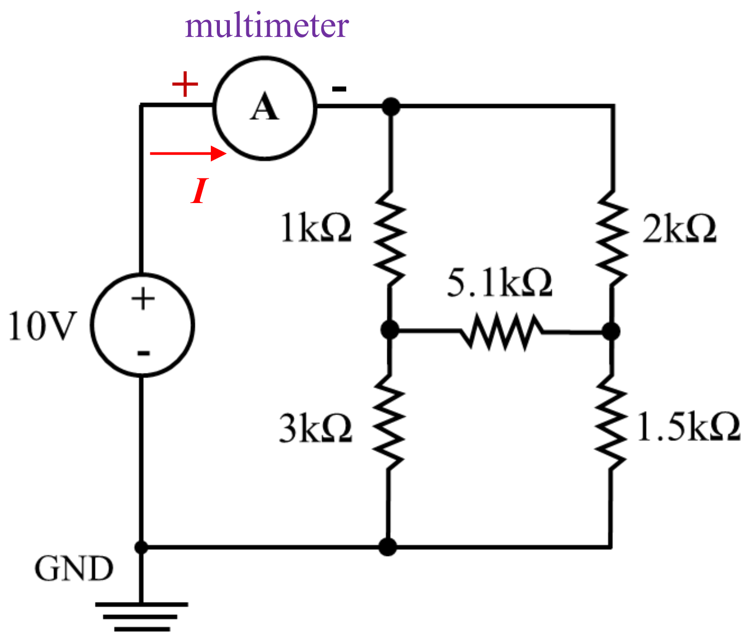
Step 3: Generate 10V from the DC power supply to the network as shown in the figure below.



Step 4: Use the multimeter "V" mode to measure the voltage at node **C** (versus GND) and the voltage at node **D** (versus GND).

Q12: Record the voltages from the multimeter and fill in the table in the summary sheet.

Step 5: Now change to use the multimeter "mA" mode to measure the total current I in the circuit. Remember to connect the multimeter in series with the resistor network as shown in the figure below.



Q13: Record the value of I from the multimeter (including the unit "mA") to fill in the table in the summary sheet.

****** TA Check 5: Demo the measurement of current I to your TA.**

Q14: Based on your record in Q13, use **Ohm's law** to calculate for R_{eq} and fill in the table in the summary sheet.

$$R_{eq} = \frac{V}{I}$$

Notice that this calculated resistance should match your measured R_{eq} in Q11.

After you have done Experiment 5, remove all of the components from your breadboard.

Remember to clean up your bench and return all the components, breadboard, tools, and cable box! A messy table will cost 1 point!